CAMPUSX • 100 DAYS OF MACHINE LEARNING

DAY 5

Complete Notes, Practical Workbook & Revision Guide

End-to-end toy project • Problem framing • Working with CSV files

Beginner-friendly English + Roman Urdu

Quiz result: 8/8 (100%) — all answers correct

Core rule: define the problem before choosing an algorithm.

1. Learning Objectives

By the end of Day 5, you should be able to frame an ML problem, identify features and target, distinguish common ML tasks, inspect CSV data with pandas, explain train/test separation, and recognize data leakage.

Roman Urdu: Pehle problem ko clearly samjho. Phir data collect aur clean karo, X aur y choose karo, model train karo, aur unseen test data par performance check karo.

2. End-to-End Machine Learning Workflow

Stage	What happens	Roman Urdu reminder
1. Problem framing	Define the goal, prediction, user, and success metric.	Asal masla aur success clear karo.
2. Data collection	Gather relevant, permitted, representative data.	Relevant data jama karo.
3. EDA and cleaning	Inspect distributions, errors, duplicates, missing values, outliers.	Data ko samjho aur issues dhoondo.
4. Feature preparation	Select and transform useful inputs when appropriate.	Inputs tayyar karo.
5. Data splitting	Keep evaluation data separate from training.	Test data alag rakho.
6. Model training	Fit an algorithm on training data.	Model ko examples se seekhne do.
7. Evaluation	Use suitable metrics and compare to a baseline.	Performance check karo.
8. Deployment/monitoring	Use the model and monitor performance and drift.	Real use mein monitor karo.

The workflow is iterative: poor results may mean you need better data, a clearer target, or improved features.

3. Framing a Problem

Ask: Who will use the prediction? What exactly must be predicted? When is the prediction needed? Which inputs are available then? What metric defines success? What are the consequences of errors?

Example: house-price prediction

Question	Example
Goal	Estimate a house's selling price before sale.
Features (X)	Area, bedrooms, age, possibly location.
Target (y)	Actual selling price.
ML task	Regression: the target is a number.
Possible metric	MAE, expressed in currency units.

Roman Urdu: Sirf 'AI model banana hai' kaafi nahi. Clear karo kya predict karna hai, kis data se, kis waqt, aur result ko kaise judge karna hai.

4. Regression, Classification, and Clustering

Task	Output	Example
Regression	A numerical value	House price, delivery time.
Classification	A class or label	Spam/not spam; fraud/not fraud.
Clustering	Groups discovered by similarity	Customer segments without predefined labels.

Rule of thumb: numerical value → often regression; known categories → often classification; discovering groups without labels → clustering. Always follow the actual problem definition.

5. Rows, Features, and Target

area	bedrooms	age	price
1000	2	5	8000000
1500	3	3	12000000
2000	4	2	17000000

Row/sample: one house. **Feature:** an input column. **Target:** the value to predict. **Dataset:** the collection of examples.

```
import pandas as pd

df = pd.read_csv("houses.csv")
X = df[["area", "bedrooms", "age"]]  # multiple input columns
y = df["price"]                      # target
print(X.head())
print(y.head())
```

Double brackets select multiple columns and return a DataFrame. Single brackets for df["price"] select one column and usually return a Series.

6. Working with CSV Files

CSV means Comma-Separated Values. It is a common text format for storing tabular rows and columns.

```
import pandas as pd

df = pd.read_csv("houses.csv")
print(df.head())          # preview first five rows
print(df.shape)           # (rows, columns)
print(df.columns)         # column names
df.info()                 # types and non-null counts
print(df.isnull().sum())  # missing count per column
print(df.describe())      # numerical summary
```

Command	Purpose	Roman Urdu
pd.read_csv(path)	Load a CSV file.	CSV load karo.
df.head()	Preview first five rows by default.	Shuru ki rows dekho.
df.shape	Returns (rows, columns).	Rows aur columns ki tadaad.
df.columns	Column names.	Columns ke naam.
df.info()	Data types and non-null counts.	Types aur filled entries.
df.isnull().sum()	Missing count per column.	Missing values count karo.
df.describe()	Summary statistics for numeric columns by default.	Numbers ka summary.

If the CSV cannot be found, check the current working directory or provide the correct path. Inspect data types and column names before modeling.

7. Training vs Testing

Training data is used to fit the model. Test data is held back to estimate performance on unseen examples. An 80/20 split is common for teaching examples, but the right split depends on the data and use case.

```
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn.metrics import mean_absolute_error

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
model = LinearRegression()
model.fit(X_train, y_train)
predictions = model.predict(X_test)
print("MAE:", mean_absolute_error(y_test, predictions))
```

MAE is the average absolute gap between actual and predicted values. It uses the same units as the target. Compare against a baseline and interpret results in context.

8. Data Leakage and Data Quality

Data leakage occurs when training/evaluation uses information that would not legitimately be available at prediction time, or when information from the test set improperly influences model training.

Examples:
• Including actual selling price in X while y is also selling price.
• Using a field created only after the event being predicted.
• Fitting preprocessing on all data before splitting when it learns statistics from the test data.

Check for missing values, duplicates, inconsistent units, impossible values, incorrect labels, and unrepresentative samples. Do not fill or delete missing data automatically without understanding the reason.

Roman Urdu: Agar model ko pehle se jawab mil jaye, to test score acha lag sakta hai lekin real life mein model fail ho sakta hai. Isay data leakage kehte hain.

9. Mini Project Plan

1. Write a clear problem statement.
2. Identify the target and candidate features.
3. Load and inspect the CSV.
4. Check data quality and plan cleaning.
5. Split training and test data.
6. Train a simple baseline model.
7. Evaluate with an appropriate metric and inspect errors.
8. Record limitations before any deployment.

10. Practice Exercises

Try before reading the answers:
1. A CSV has 1,000 houses and 5 columns. What is df.shape?
2. If price is the target, what belongs in X and y?
3. Fraudulent/legitimate prediction: regression or classification?
4. Explain training vs test data.
5. Write code to load data.csv, preview rows, and count missing values.
6. Why should price not be included in X if y is price?
7. What does clustering do without labels?
8. Name two data-quality problems.

11. Answer Key

1. (1000, 5).
2. y = df["price"]; X contains legitimate input columns such as area, bedrooms, and age, not price.
3. Classification, because the output is a category.
4. Training fits the model; test data estimates performance on unseen examples.
5. df = pd.read_csv("data.csv"); print(df.head()); print(df.isnull().sum()).
6. It gives the answer to the model as an input and can cause leakage.
7. It discovers groups of similar examples.
8. Missing values, duplicates, incorrect types, outliers, inconsistent units, or wrong labels.

12. Your Day 5 Quiz Review

Score: 8/8 (100%). Every submitted answer was correct; no quiz mistakes were found.

Concept	Your answer	Result
House-price task	Regression	Correct
Target	Price	Correct
df.shape	Rows and columns	Correct
Read CSV	pd.read_csv()	Correct
Why test data?	Evaluate on unseen data	Correct
Missing-value count	df.isnull().sum()	Correct
Groups without labels	Clustering	Correct
Data leakage	Using information that should not be available at prediction time	Correct

13. Revision Checklist

- ■ I can explain the end-to-end workflow.
- ■ I can define goal, target, features, and success metric.
- ■ I can distinguish regression, classification, and clustering.
- ■ I understand rows, X, and y.
- ■ I can load and inspect a CSV using pandas.
- ■ I can interpret df.shape and count missing values.
- ■ I can explain why test data is held back.
- ■ I can give examples of data leakage.

14. Roman Urdu Final Recap

Day 5 mein hum ne seekha ke ML project se pehle problem define karna zaroori hai. Regression number predict karta hai, classification category predict karta hai, aur clustering similar examples ke groups banati hai. CSV ko pandas se read karke head(), shape, info(), isnull().sum(), aur describe() se inspect kar sakte hain. Features X hotay hain aur target y hota hai. Training data model ko sikhata hai; test data unseen examples par performance check karta hai. Data leakage se bachna bohat zaroori hai.

Next step: Day 6 par tab move karein jab aap workflow explain kar sakein aur practice script khud run kar lein.